Aerodynamic Properties of the Wing Washout Design

Project Report

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Introduction:

Our Science Fair project is called “Aerodynamic Properties of the Wing Washout Design”. To investigate the influence of wing twist on the aerodynamic performance of a wing, a Boeing 737 wing model was utilized. Variants with twist angles of 0°, 3°, and 5° relative to the baseline reference plane were created by adjusting the wing root twist angle. Computational Fluid Dynamics (CFD) simulations were conducted to evaluate the lift, drag, lift coefficient, drag coefficient, and lift-to-drag ratio across various angles of attack (0°, 3°, 6°, 9°, 12°, 15°, and 18°) at a Mach number of 0.6. We conjecture that a higher washout angle will lead to higher drag near the 0 angle of attack, but it will lead to higher lift at an angle of attack near the critical angle.

The washout design, also called a wing twist, is a common design feature in aircraft wings, employed to optimize aerodynamic performance by reducing induced drag, improving lift distribution, and preventing wingtip stalls. Wing twists can be categorized into geometric twists and aerodynamic twists. In a geometric twist, the wingtip section is typically twisted downward relative to the wing root, reducing the angle of attack at the wingtips. This design reduces aerodynamic separation on wing tip surfaces. In aerodynamic twists, the wing sections lie in the same geometric plane, but asymmetric airfoil profiles with varying curvature are used along the span. This approach modifies the aerodynamic characteristics of the wing, achieving an effect similar to a geometric twist in controlling lift distribution. Both forms of twist play a critical role in optimizing wing performance and ensuring safer flight characteristics.

Our research will provide empirical data on the relationship between lift, drag, and the washout degree. Airplane designers could utilize this result to determine the washout angle that best balances their requirements for safety and efficiency.

Materials and procedures:

Theoretical background:

This study utilizes some important definitions, including the lift force, drag force, lift coefficient, drag coefficient, angle of attack, washout angle, and the Reynolds Number (Re). Their definitions and meanings are as follows:

Lift force is a force on a body in a direction normal to the flow direction. The lift coefficient (C_L) is a dimensionless coefficient that relates the lift force, velocity, and wing area. The equation is shown in the below equation:

$$C_L = \frac{F_L}{1/2\rho V^2 A}$$

Drag force is a force on a body parallel to the incoming flow. Similar to the C_L, the drag coefficient relates the drag force, velocity, and wing area. The equation is shown below:

$$C_D = \frac{F_D}{1/2\rho V^2 A}$$

The Reynolds number (R_e) is another important dimensionless quantity that predicts the flow pattern of a fluid by measuring the ratio between inertial forces and viscous forces. The equation is shown below:

$$R_e = \frac{\rho V c}{\mu}$$

The angle of attack is the angle between the chord line and the incoming flow, which is shown in Figure 1.



Figure 1. Angle of attack (Angle of Attack, 2024)

The washout angle describes the difference in the angle of incidence between the wing root and the wing tip. Essentially, the higher the washout angle, the more “twisted” the wing is. The diagram is shown in Figure 2.

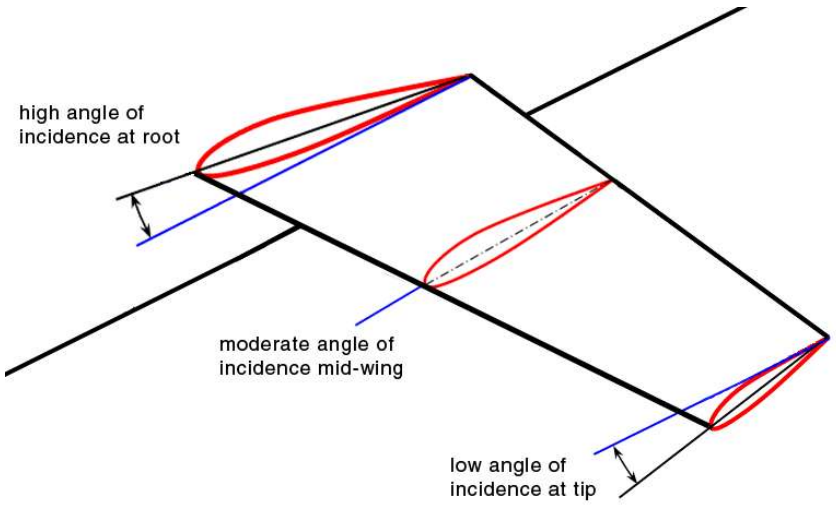


Figure 2. Washout angle (Washout, 2024)

Our study focuses on the aerodynamic characteristics of the wing flow field under varying twist angles and angles of attack. By constructing a wing model and simulating the surrounding flow field under different operating conditions, the aerodynamic properties of the flow field in each scenario are analyzed. The original geometric characteristics of the wing are illustrated in Figure 3. The wing comprises two primary airfoils and four characteristic airfoils. Airfoils 1 and 2 are symmetrical, while airfoils 3 and 4 are asymmetric, with differing curvature profiles. The root chord length is 4.9 m, the tip chord length is 1.25 m, the half-span length is 10.44 m, and the quarter-chord sweep angle is 25°.

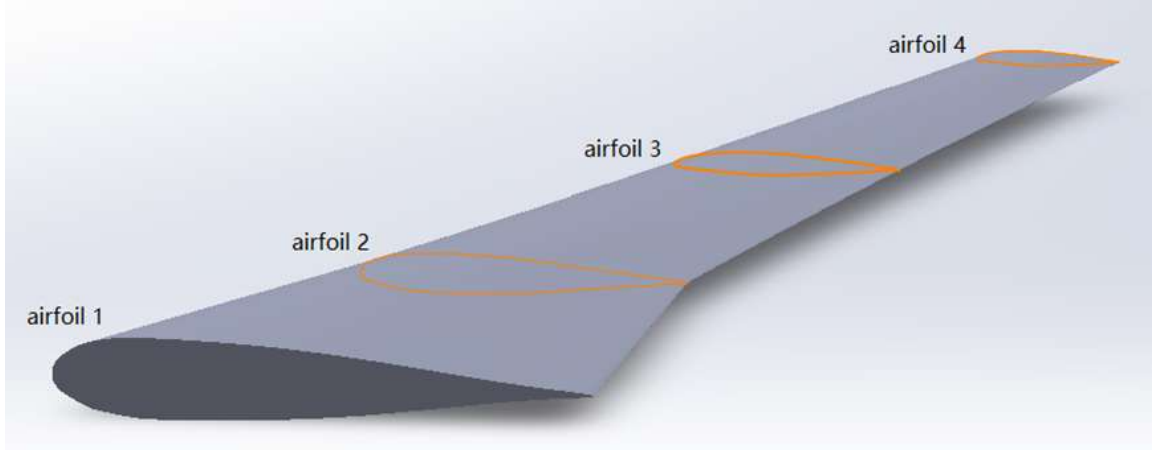


Figure 3. Original Geometric Characteristics of the Wing

When the wing is subjected to twisting, the plane of Airfoil 4 rotates downward relative to the plane of Airfoil 3. This study examines twist angles of 0°, 3°, and 5°. The 3° and 5° wing twist configurations are depicted in Figures 4 and 5, respectively. Based on these twist configurations, models with varying angles of attack are developed for further analysis.

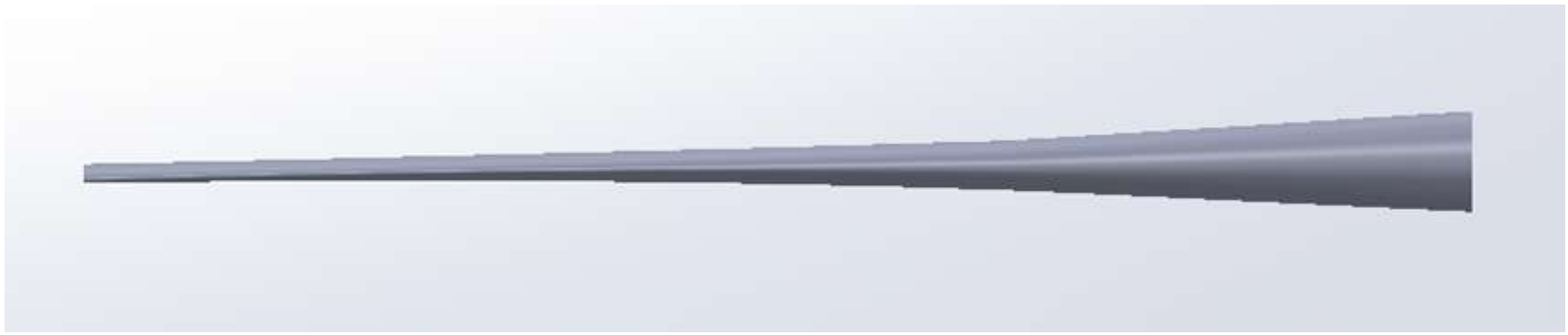


Figure 4: 3° washout angle model

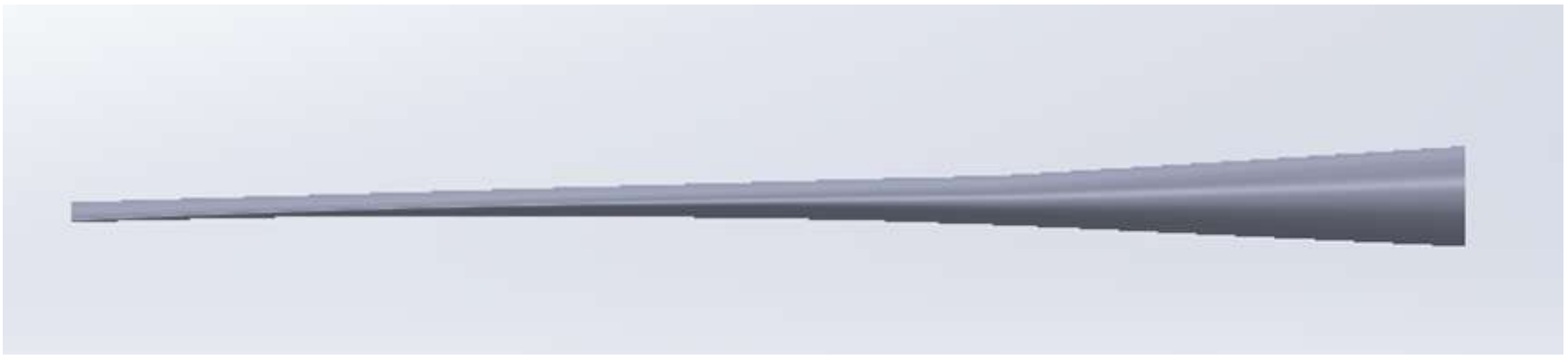
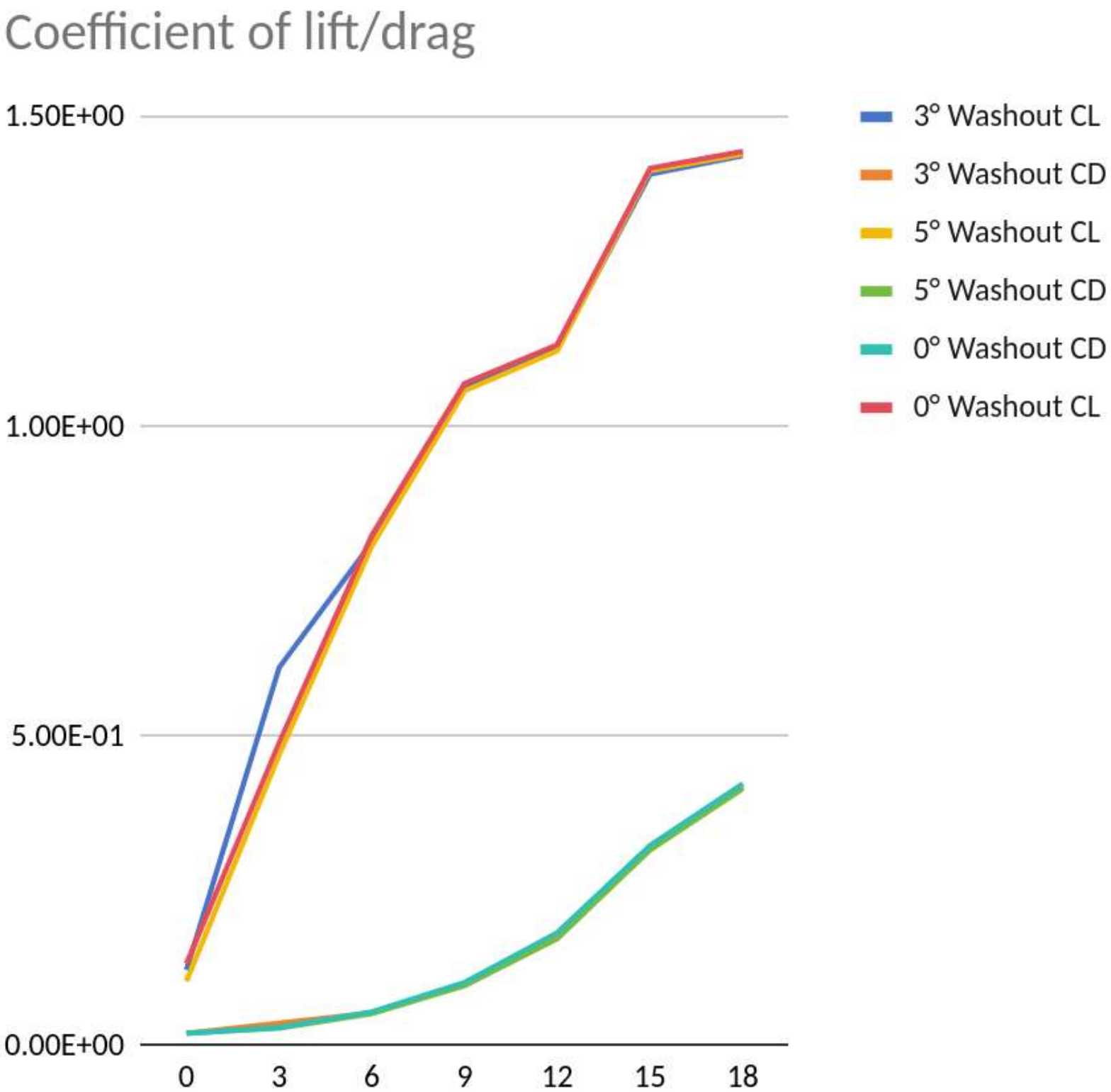


Figure 5: 5° washout angle model

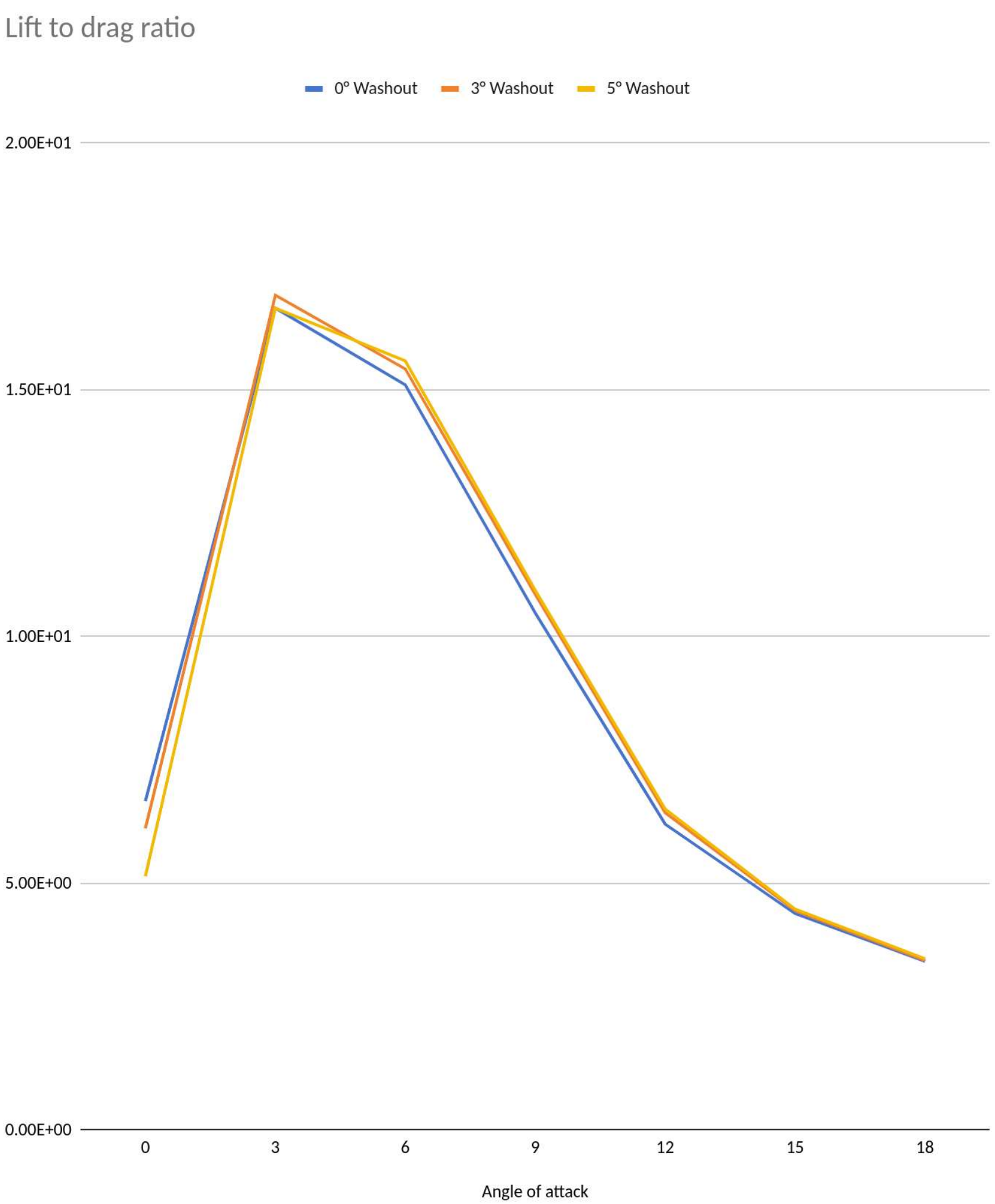
Data Collected:

Airfoil-Washout angle -Angle of attack	Lift Force	Coefficient of Lift	Drag Force	Coefficient of drag	**Lift-to-drag ratio**
737-0-0	9.49E+04	1.32E-01	1.42E+04	1.99E-02	**6.66E+00**
737-0-3	3.51E+05	4.89E-01	2.11E+04	2.94E-02	**1.67E+01**
737-0-6	5.91E+05	8.24E-01	3.91E+04	5.46E-02	**1.51E+01**
737-0-9	7.67E+05	1.07E+00	7.32E+04	1.02E-01	**1.05E+01**
737-0-12	8.12E+05	1.13E+00	1.31E+05	1.83E-01	**6.20E+00**
737-0-15	1.02E+06	1.42E+00	2.32E+05	3.23E-01	**4.39E+00**
737-0-18	1.04E+06	1.44E+00	3.03E+05	4.23E-01	**3.42E+00**
737-3-0	8.73E+04	1.22E-01	1.43E+04	1.99E-02	**6.11E+00**
737-3-3	4.38E+05	6.11E-01	2.59E+04	3.61E-02	**1.69E+01**
737-3-6	5.83E+05	8.13E-01	3.78E+04	5.27E-02	**1.54E+01**
737-3-9	7.64E+05	1.07E+00	7.04E+04	9.81E-02	**1.09E+01**
737-3-12	8.07E+05	1.13E+00	1.26E+05	1.75E-01	**6.43E+00**
737-3-15	1.01E+06	1.41E+00	2.27E+05	3.17E-01	**4.44E+00**
737-3-18	1.03E+06	1.44E+00	3.00E+05	4.18E-01	**3.44E+00**
737-5-0	7.45E+04	1.04E-01	1.45E+04	2.02E-02	**5.14E+00**
737-5-3	3.37E+05	4.69E-01	2.02E+04	2.82E-02	**1.67E+01**
737-5-6	5.79E+05	8.07E-01	3.71E+04	5.18E-02	**1.56E+01**
737-5-9	7.58E+05	1.06E+00	6.93E+04	9.66E-02	**1.09E+01**
737-5-12	8.05E+05	1.12E+00	1.24E+05	1.72E-01	**6.50E+00**
737-5-15	1.01E+06	1.41E+00	2.26E+05	3.16E-01	**4.48E+00**
737-5-18	1.03E+06	1.44E+00	2.98E+05	4.15E-01	**3.47E+00**

Table 1: Lift and drag of the washout wing as the angle of attack changes



Graph 1: Coefficient of lift/drag vs. Angle of attack graph



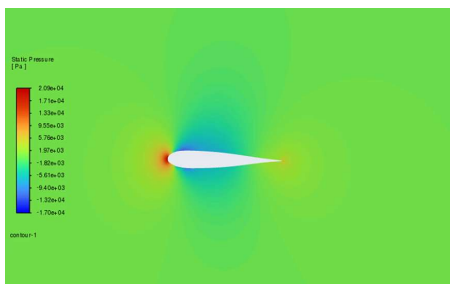
Graph 2: Lift-to-drag ratio vs. Angle of attack graph

Results/Data Analysis:

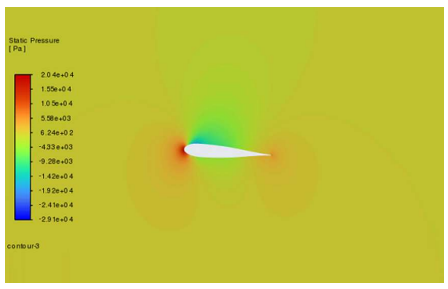
As we can see in table 1, graph 1, and graph 2, they show the wing's lift/drag and lift-to-drag ratio as the angle of attack changes. As the angle of attack increases, lift increases, and drag also increases, but the lift-to-drag ratio increases first and then decreases as the angle of attack increases. When the angle of attack is less than 3°, the lift-to-drag ratio increases rapidly, reaching a maximum value at around 3°. After the angle of attack is greater than 6°, the lift-to-drag ratio gradually decreases as the angle of attack increases. After the wingtip is washed outward for 5°, the lift and drag at the same angle of attack decrease compared to the case where the wingtip is not washed outward, but the lift-to-drag ratio increases slightly. The lift-to-drag ratio of the wing is the best at an angle of attack of 3°~6°.

Wing root:

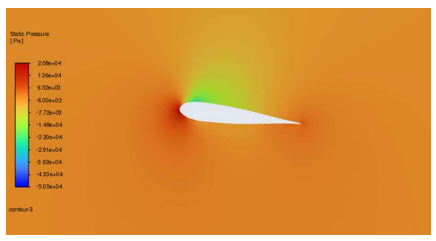
Pressure Cloud Map of 0° washout angle (at 0m to the symmetrical plane):



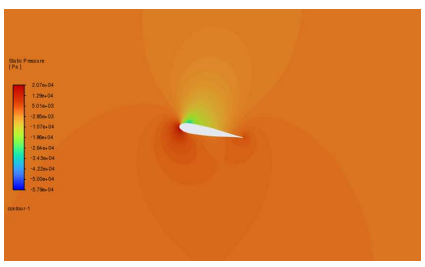
0° angle of attack ↑



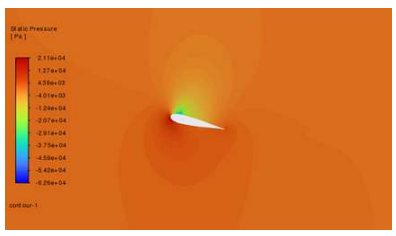
3° angle of attack ↑



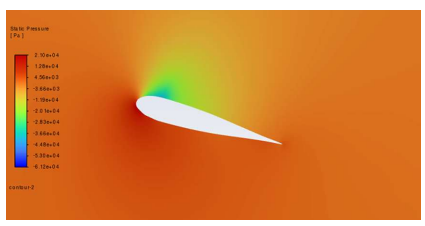
6° angle of attack ↑



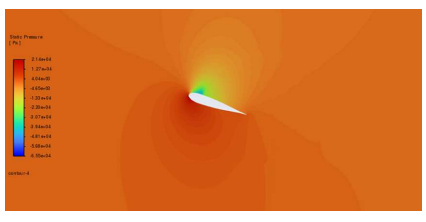
9° angle of attack ↑



12° angle of attack ↑



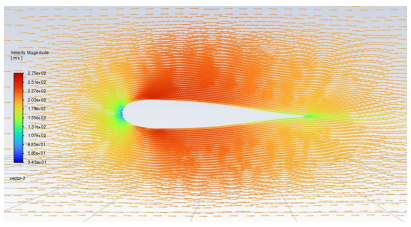
15° angle of attack ↑



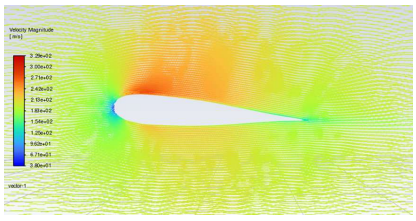
18° angle of attack ↑

Analysis: The pressure on the lower wing surface of the aircraft is greater than that on the upper wing surface. As the angle of attack increases, the high-pressure airflow below rolls upward along the wing tip to the low-pressure area above. The adverse pressure gradient on the upper surface increases and the pressure difference resistance gradually increases.

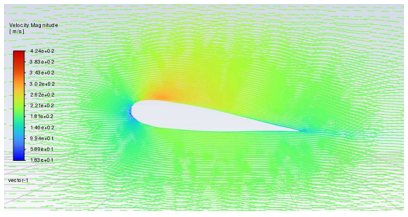
Velocity Vectors of 0° washout angle:



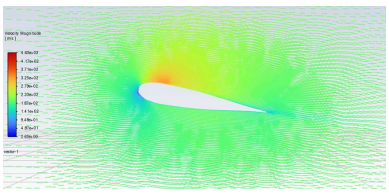
0° angle of attack ↑



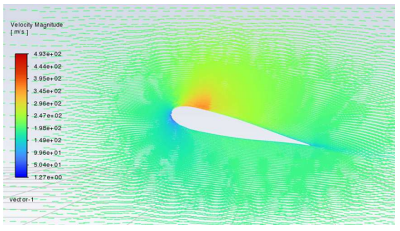
3° angle of attack ↑



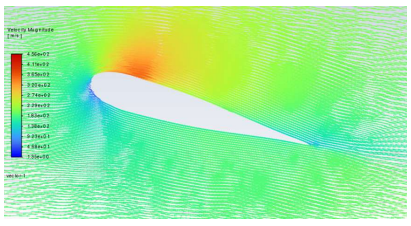
6° angle of attack ↑



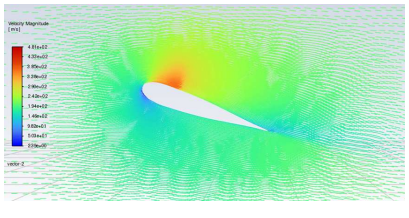
9° angle of attack ↑



12° angle of attack ↑



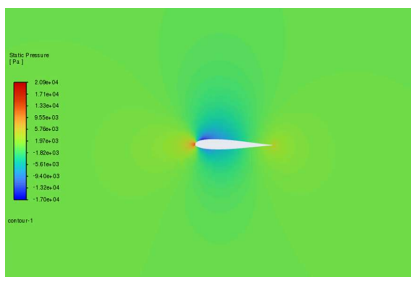
15° angle of attack ↑



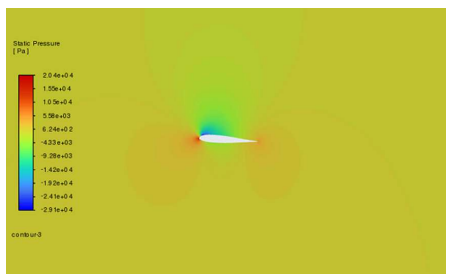
18° angle of attack ↑

Analysis: At 0° angle of attack, the symmetric wing didn't lead to a significant difference between the velocity of the air current on the upper edge and the lower edge of the wing. As the angle of attack increases, the difference in velocity between the airflow between the upper edge and bottom edge increases, with the upper edge having a larger velocity. Due to Bernoulli's law, the wing now generates lift. Additionally, the boundary layer separation isn't significant in all of the angles of attacks tested, so the wing hasn't stalled.

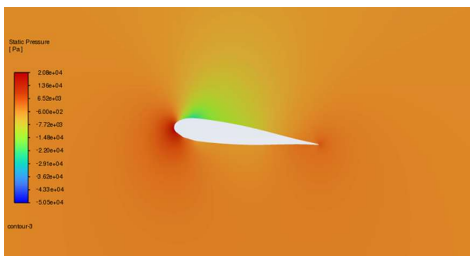
Pressure Cloud Map at 3m to symmetrical plane (0° washout angle):



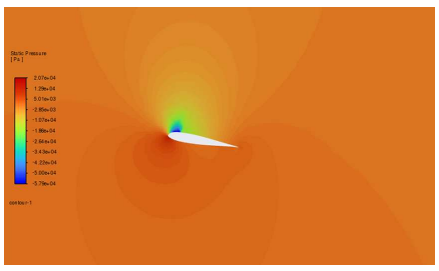
0° angle of attack ↑



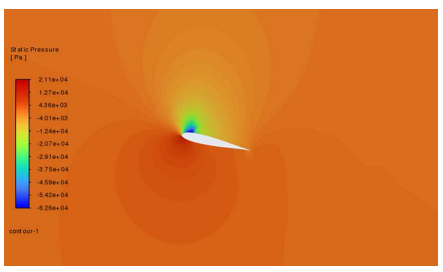
3° angle of attack ↑



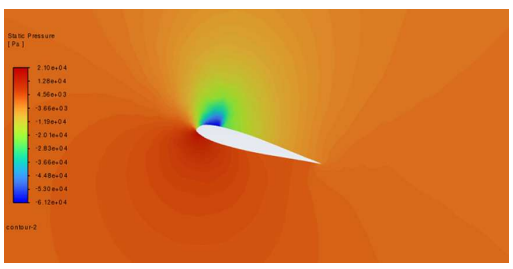
6° angle of attack ↑



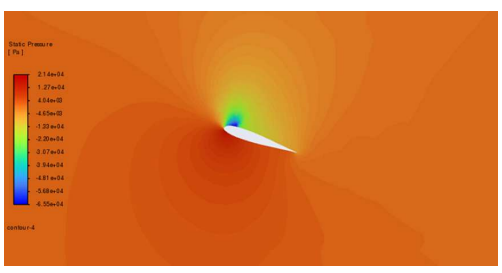
9° angle of attack ↑



12° angle of attack ↑



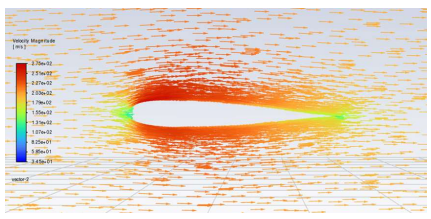
15° angle of attack ↑



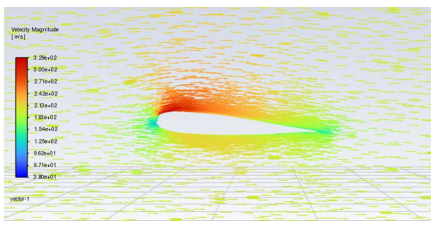
18° angle of attack ↑

Analysis: Similar to the data of the wing root, the pressure pattern 3m to the symmetrical plane increases and then decreases. The pressure difference increases as the angle of attack increases from 0o to 15o, and then the pressure on the top of the airfoil starts to decline near an angle of attack of 18°.

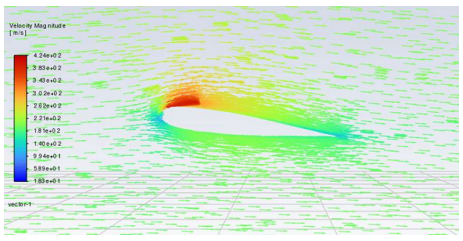
Velocity Vectors at 3m to symmetrical plane (0° washout angle):



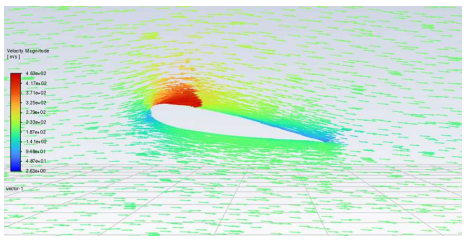
0° angle of attack ↑



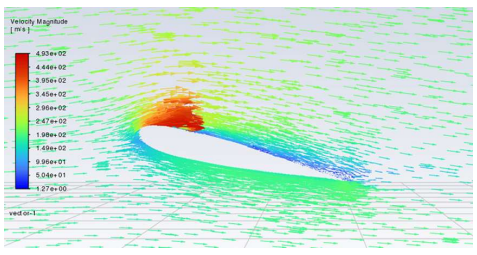
3° angle of attack ↑



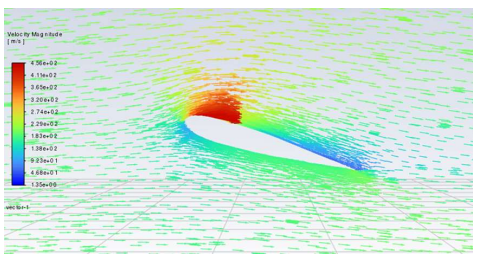
6° angle of attack ↑



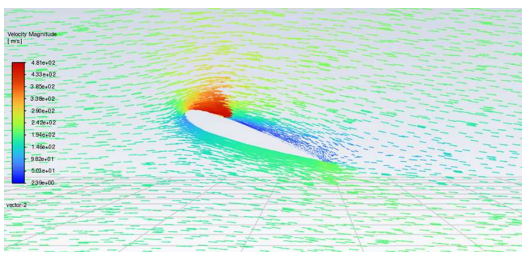
9° angle of attack ↑



12° angle of attack ↑



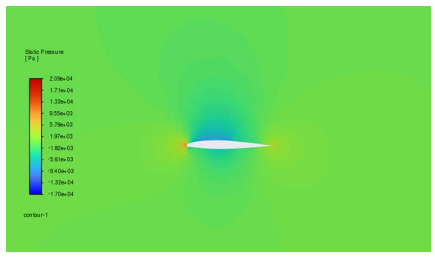
15° angle of attack ↑



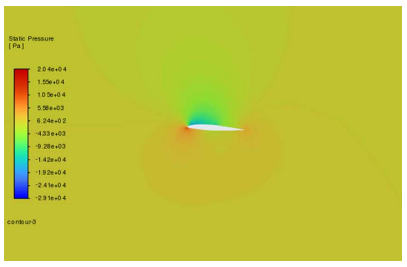
18° angle of attack ↑

Analysis: At low angles of attack, there is no flow separation before the trailing edge, and the flow remains attached to the rear portion of the wing's upper surface as a turbulent boundary layer. As the angle of attack increases, the adverse pressure gradient on the upper surface intensifies, leading to a more rapid reduction in surface kinetic energy. This causes flow separation to initiate near the trailing edge, resulting in a gradual increase in pressure drag. With a further increase in the angle of attack, the separation point shifts forward, reducing the chord-wise region capable of generating lift.

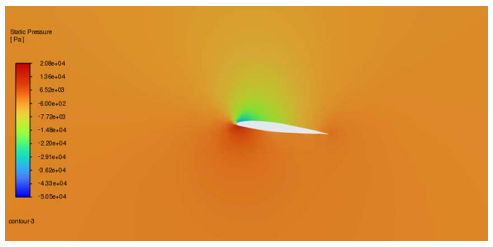
Pressure Cloud Map at 7m to symmetrical plane (0° washout angle):



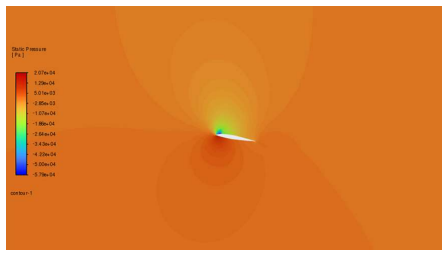
0° angle of attack ↑



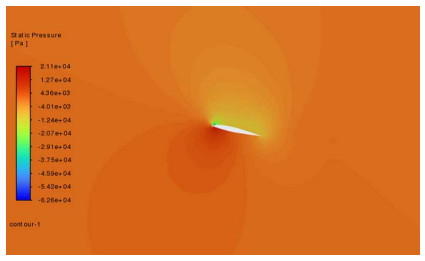
3° angle of attack ↑



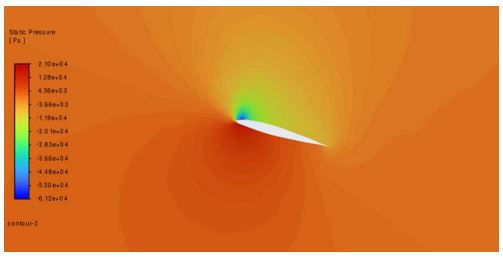
6° angle of attack ↑



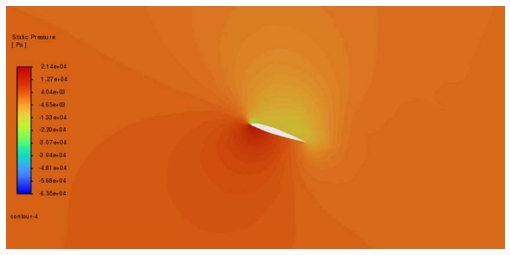
9° angle of attack ↑



12° angle of attack ↑



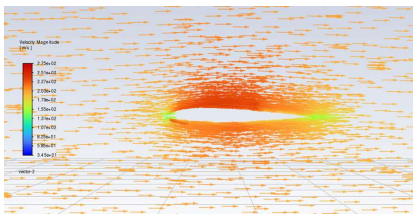
15° angle of attack ↑



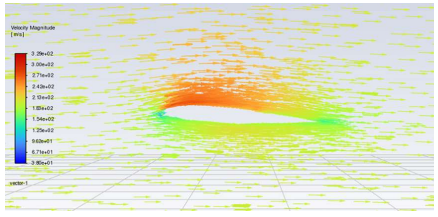
18° angle of attack ↑

Analysis: As the angle of attack increases from 0º to 15º, the pressure difference between the top and the bottom of the airfoil increases, with the upper edge having a higher pressure. This implies that the overall lift is increasing. However, when the angle of attack is 16 degrees which is the critical stall angle, the lift is maximum at 16 degrees, and then the lift decreases as the angle of attack increases.

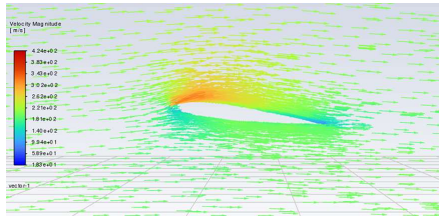
Velocity Vectors at 7m to symmetrical plane (0° washout angle):



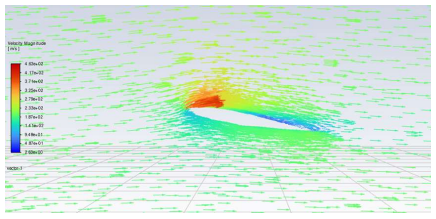
0° angle of attack ↑



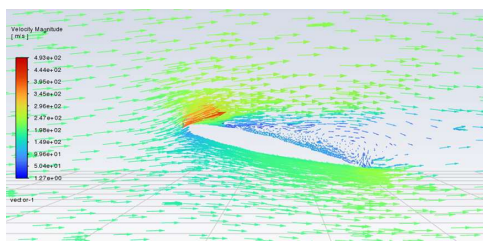
3° angle of attack ↑



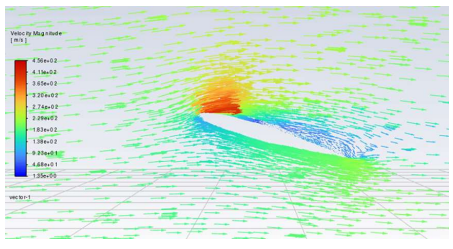
6° angle of attack ↑



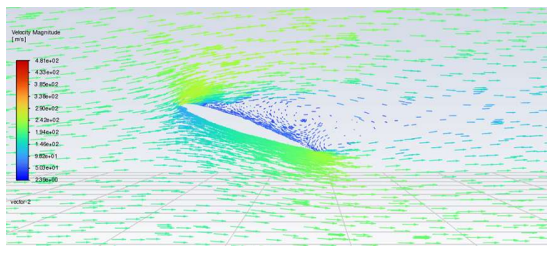
9° angle of attack ↑



12° angle of attack ↑



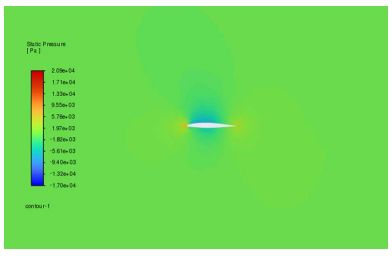
15° angle of attack ↑



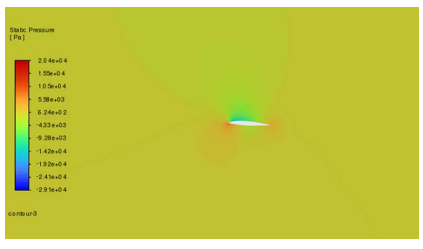
18° angle of attack ↑

Analysis: The velocity at the top edge increases while the velocity at the bottom decreases as the angle of attack increases from 0° to 15°. When the angle of attack is greater than 15°, the boundary layer separation and the turbulent flow are extremely severe, as shown by a large amount of blue velocity vector at the top of the airfoil. So, at this angle of attack, the airfoil has stalled.

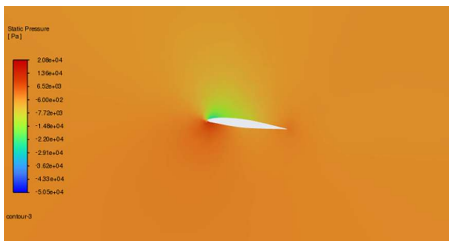
Pressure Cloud Map at 10m to symmetrical plane (0° washout angle):



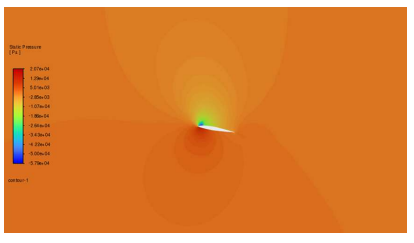
0° angle of attack ↑



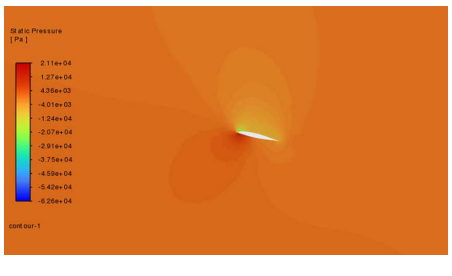
3° angle of attack ↑



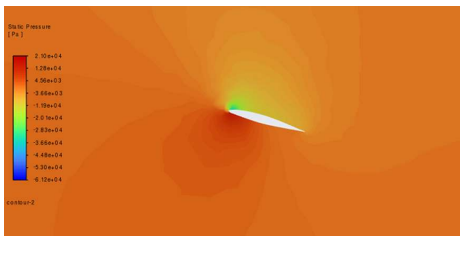
6° angle of attack ↑



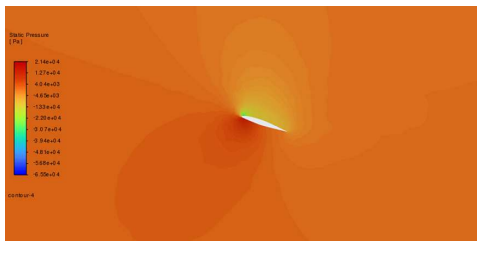
9° angle of attack ↑



12° angle of attack ↑



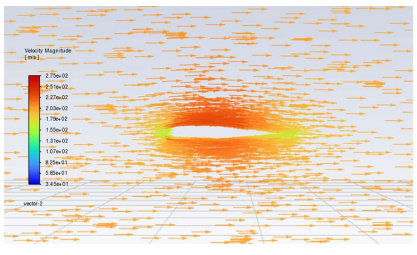
15° angle of attack ↑



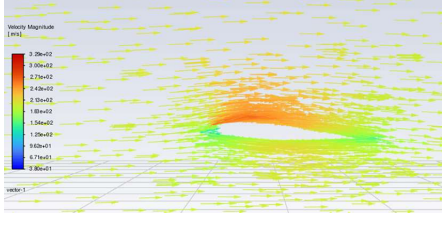
18° angle of attack ↑

Analysis: As the angle attack increases from 0° to 9°, the pressure difference between the top and the bottom of the airfoil increases, resulting in higher lift. The angle of attack keeps increasing, and the difference in pressure increases quickly, which shows an increasing lift.

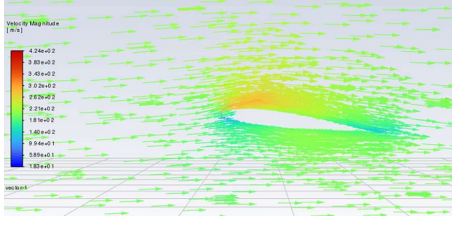
Velocity Vectors at 10m to symmetrical plane (0° washout angle):



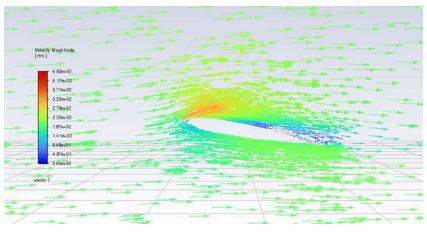
0° angle of attack ↑



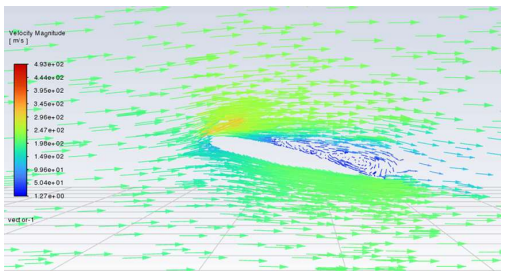
3° angle of attack ↑



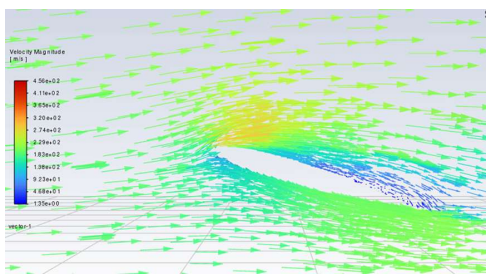
6° angle of attack ↑



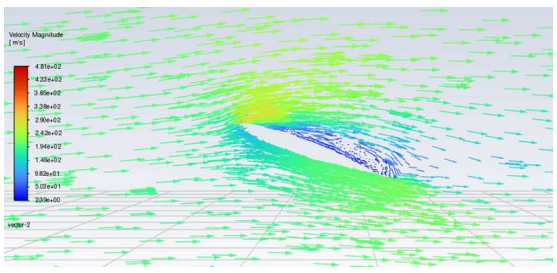
9° angle of attack ↑



12° angle of attack ↑



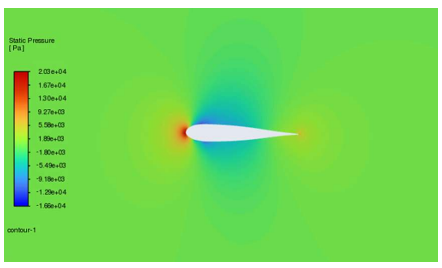
15° angle of attack ↑



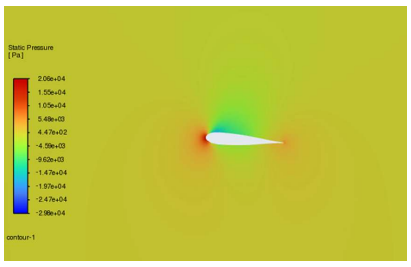
18° angle of attack ↑

Analysis: The velocity of the airflow on the top of the airfoil slowly increases as the angle of attack increases from 0° to 9°. After the angle of attack exceeds 9°, the boundary layer at the top of the airfoil separates, creating a region with turbulent flow and low velocity, thus airflow separation is severe.

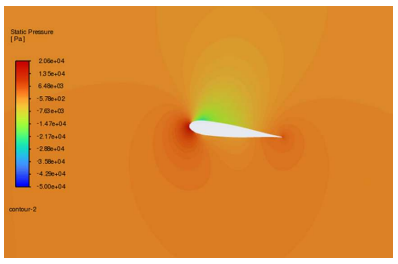
Pressure Cloud Map of 5° washout angle (at 0m to the symmetrical plane):



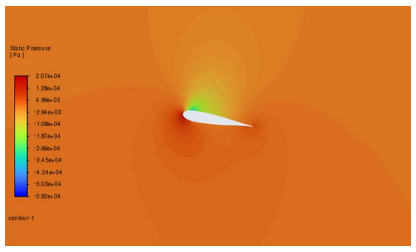
0° angle of attack



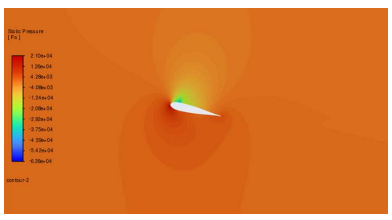
3° angle of attack ↑



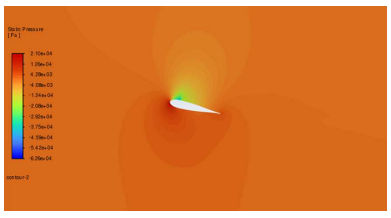
6° angle of attack ↑



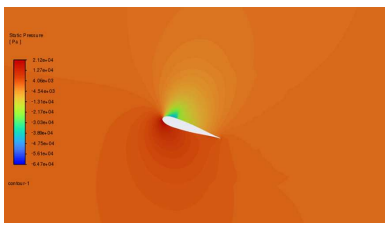
9° angle of attack ↑



12° angle of attack ↑



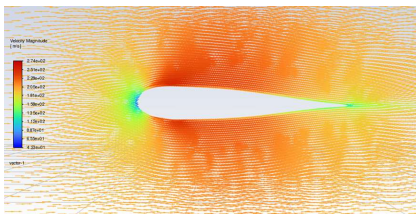
15° angle of attack ↑



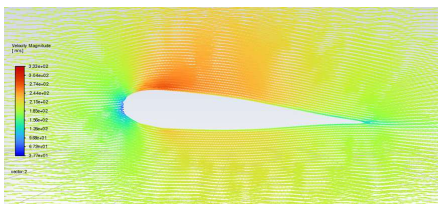
18° angle of attack ↑

Analysis: When the washout angle is 5°, at 0m from the symmetrical plane, the pressure cloud map shows a negative pressure above the airfoil and a positive pressure below the airfoil. The pressure difference thus leads to lift. Also, the difference in pressure increases as the angle of attack increases, which results in an increasing lift.

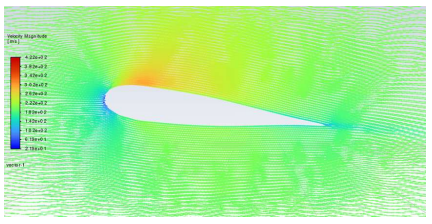
Velocity Vectors of 5° washout angle (at 0m to the symmetrical plane):



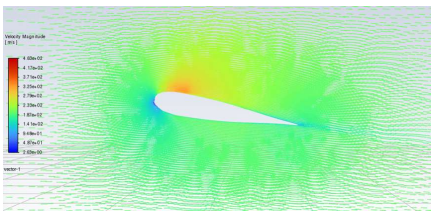
0° angle of attack ↑



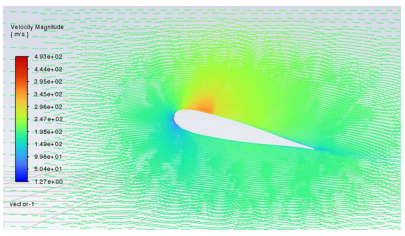
3° angle of attack ↑



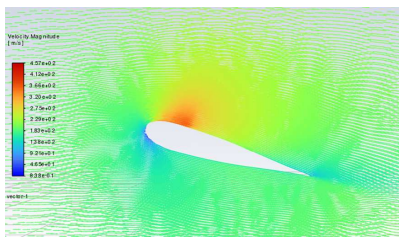
6° angle of attack ↑



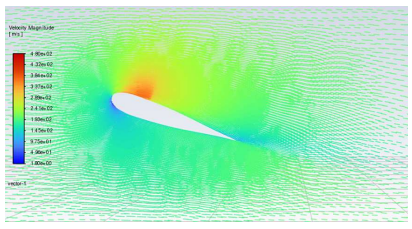
9° angle of attack ↑



12° angle of attack ↑



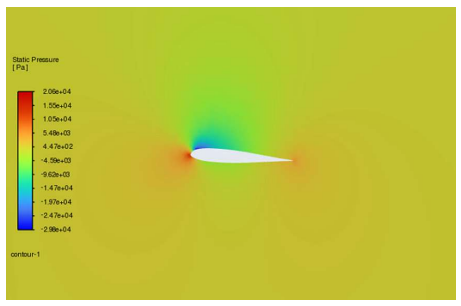
15° angle of attack ↑



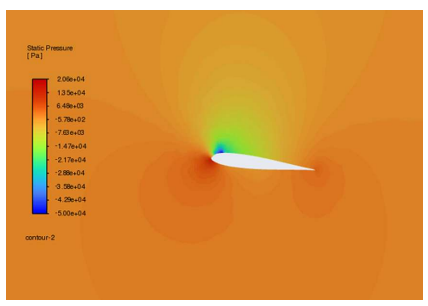
18° angle of attack ↑

Analysis: For an airfoil with a 5° washout at 0m to the symmetrical plane, the boundary layer showed little to no signs of separation across all the angles of attacks tested (0°-18°). Also, the velocity on the top of the airfoil is greater than the bottom, which leads to lower pressure on the top, thus creating lift.

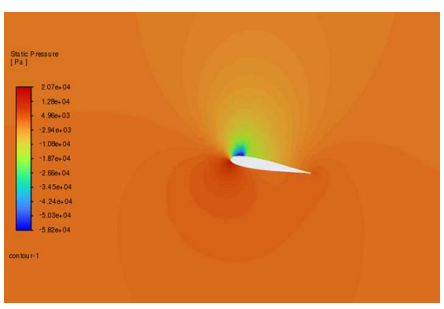
Pressure Cloud Map at 3m to symmetrical plane (5° washout angle):



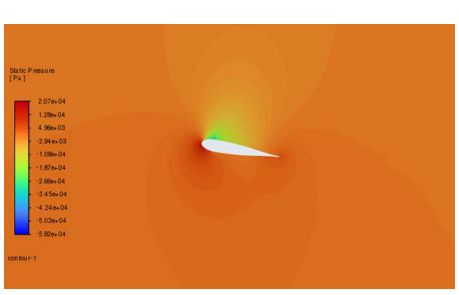
0° angle of attack ↑



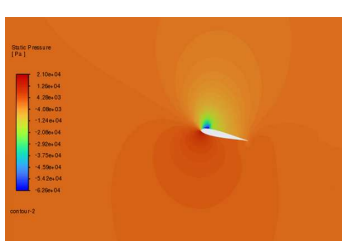
3° angle of attack ↑

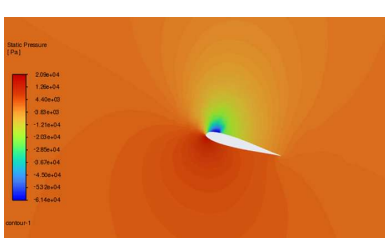


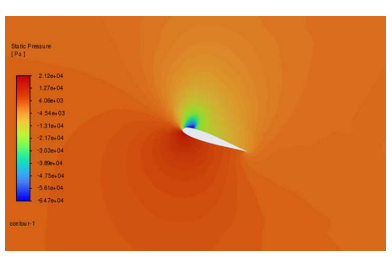
6° angle of attack ↑



9° angle of attack ↑



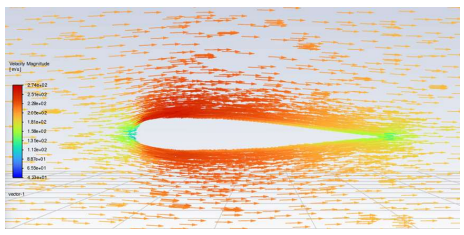




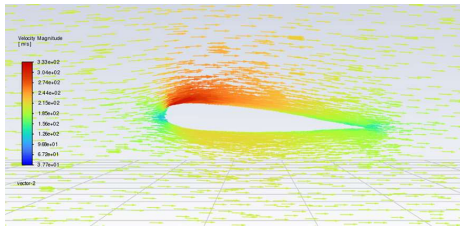
12° angle of attack ↑ 15° angle of attack ↑ 18° angle of attack ↑

Analysis: For an airfoil with a 5° washout 3m from the symmetrical plane, the pressure at the front top section of the airfoil decreases while the angle of attack increases, while the pressure at the bottom section increases while the angle of attack increases. These combine to form a lift force that increases as the angle of attack increases.

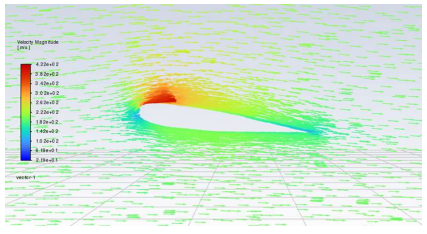
Velocity Vectors at 3m to symmetrical plane (5° washout angle):



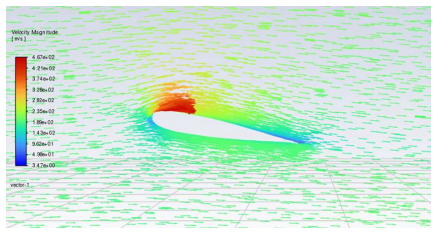
0° angle of attack ↑



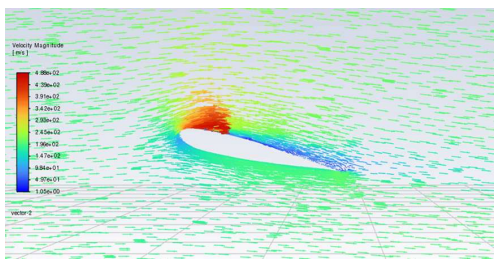
3° angle of attack ↑



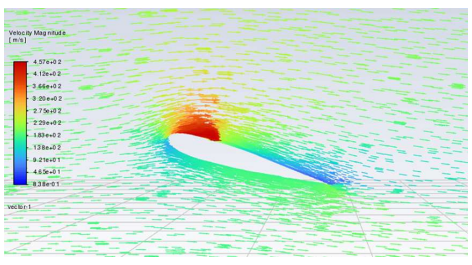
6° angle of attack ↑



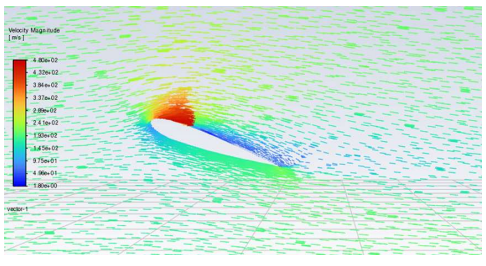
9° angle of attack ↑



12° angle of attack ↑



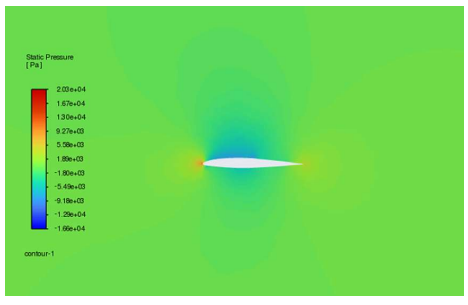
15° angle of attack ↑



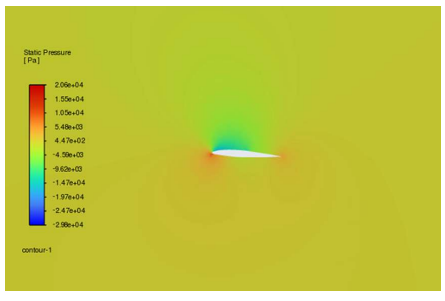
18° angle of attack ↑

Analysis: As the angle of attack increases, the velocity of the fluent at the front edge of the airfoil increases, while that at the bottom of the airfoil decreases. After an angle of attack of 12°, the boundary layer starts to separate, leading to an area with turbulent flow. However, the separation is not severe.

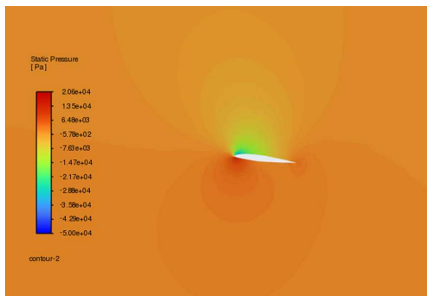
Pressure Cloud Map at 7m to symmetrical plane (5° washout angle):



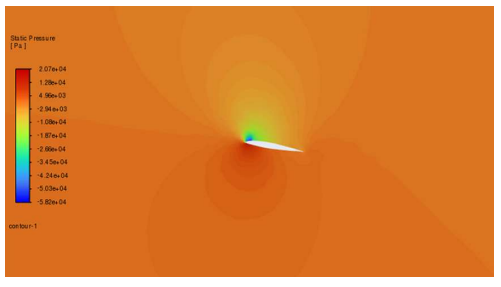
0° angle of attack ↑



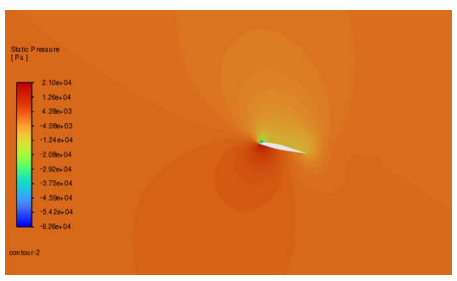
3° angle of attack ↑



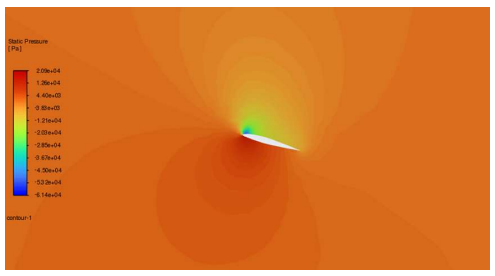
6° angle of attack ↑



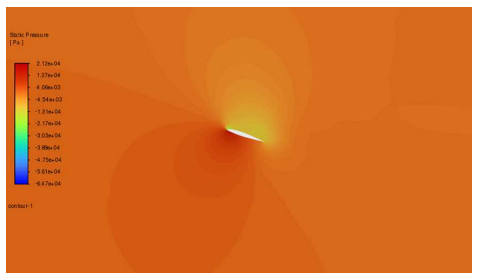
9° angle of attack ↑



12° angle of attack ↑



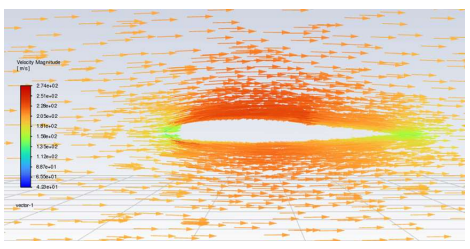
15° angle of attack ↑



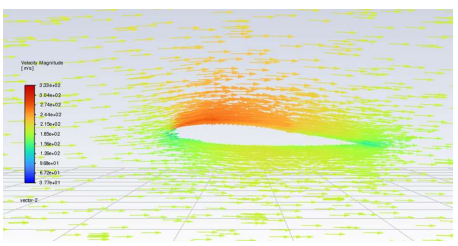
18° angle of attack ↑

Analysis: The pressure at the top of the airfoil decreases while the pressure at the bottom of the airfoil increases as the angle of attack increases from 0°-15°. After the angle of attack exceeds 15°, the change in pressure is insignificant.

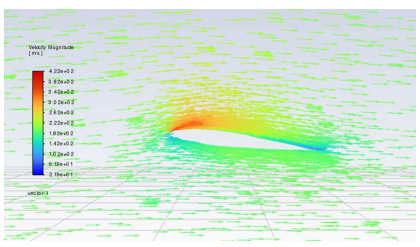
Velocity Vectors at 7m to symmetrical plane (5° washout angle):



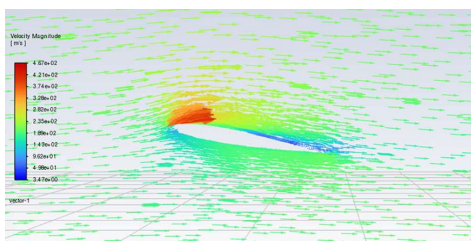
0° angle of attack ↑



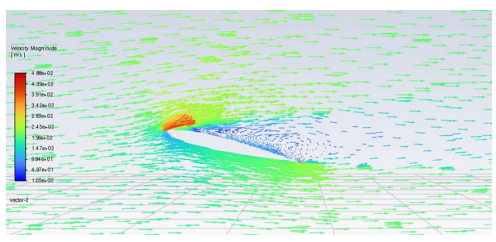
3° angle of attack ↑



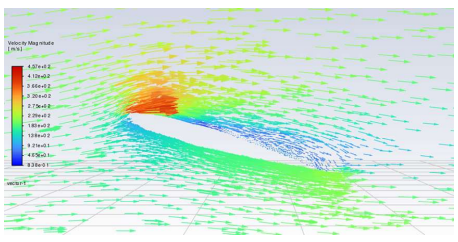
6° angle of attack ↑



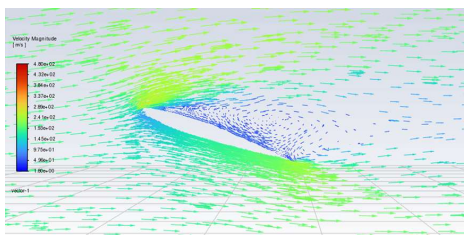
9° angle of attack ↑



12° angle of attack ↑



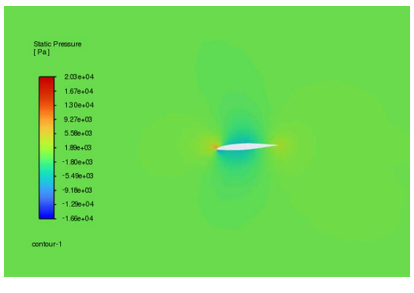
15° angle of attack ↑



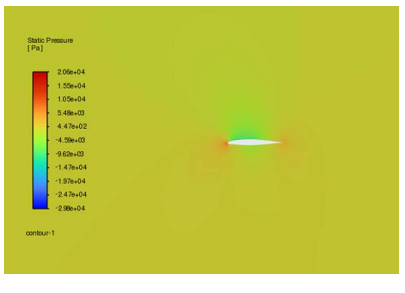
18° angle of attack ↑

Analysis: General trends are the same as previous airfoils tested. However, boundary layer separation occurs when the angle of attack is 9°, and it covers 50% of the wing. At 12°, a larger vortex is formed, and the separation surface moves forward as the angle of attack increases.

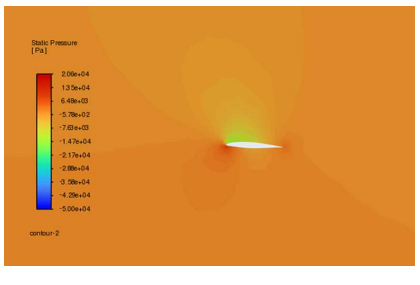
Pressure Cloud Map at 10m to symmetrical plane (5° washout angle):



0° angle of attack ↑



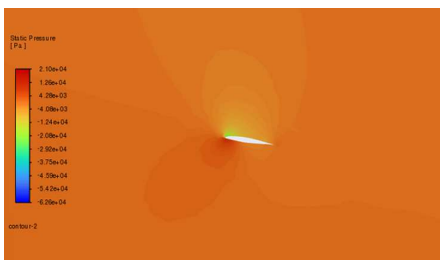
3° angle of attack ↑



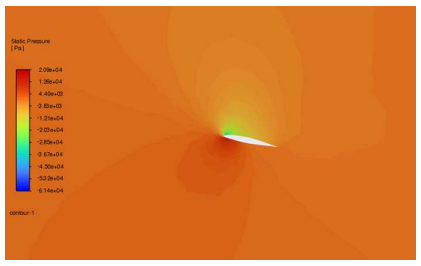
6° angle of attack ↑



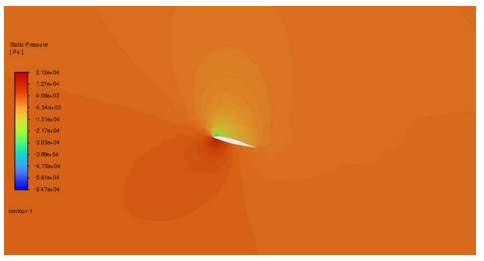
9° angle of attack ↑



12° angle of attack ↑



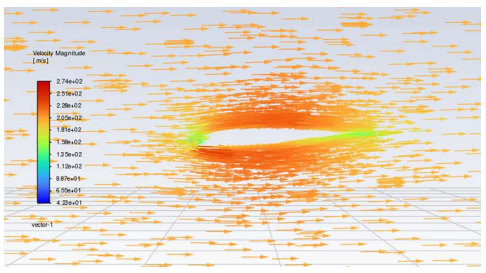
15° angle of attack ↑



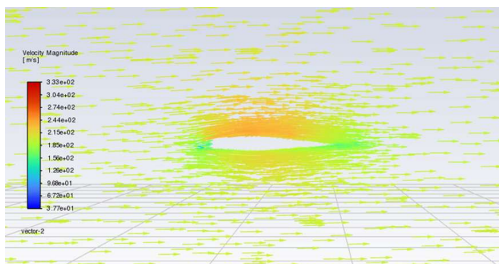
18° angle of attack ↑

Analysis: General trends of pressure are the same as previous airfoils.

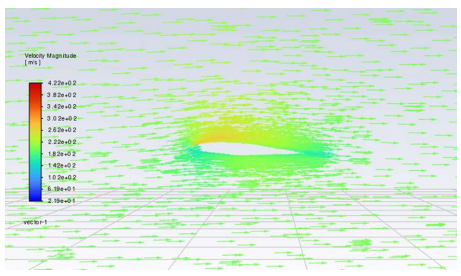
Velocity Vectors at 10m to symmetrical plane (5° washout angle):



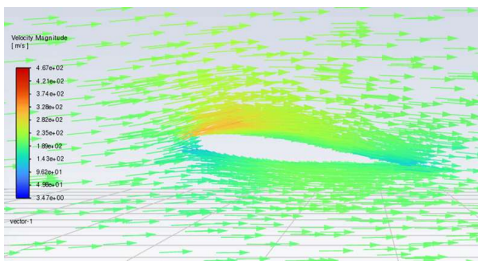
0° angle of attack ↑



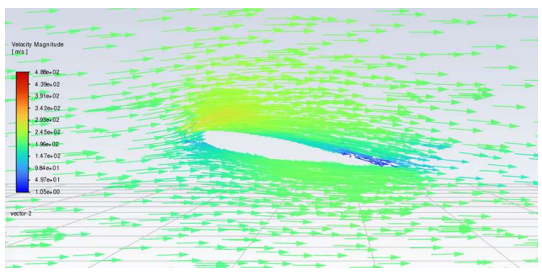
3° angle of attack ↑



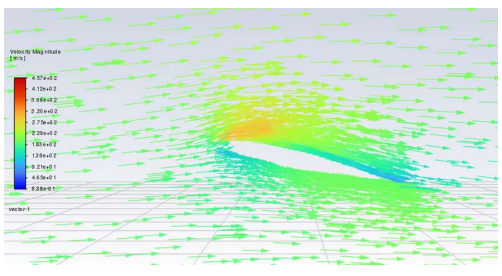
6° angle of attack ↑



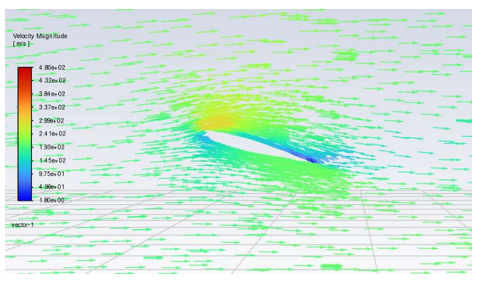
9° angle of attack ↑



12° angle of attack ↑



15° angle of attack ↑



18° angle of attack ↑

Analysis: General trends of velocity are the same as previous airfoils. However, the difference in velocity between the fluent on the top and bottom of the airfoil is less significant. Additionally, although boundary layer separation occurs at an angle of attack of 12°, it only affects a relatively small portion of the airfoil, and no complete separation has happened even as the angle of attack increases to 18°. This shows the significant effect of the washout angle.

Discussion:

Important trends are shown across the data collected. As the angle attack increases, the boundary layer on the top and rear section of the airfoil starts to separate from the airfoil, which creates an area with turbulent flow and high pressure, though both lift and drag are increasing, their ratio increases a little and starts to decrease dramatically. According to our data and research, when the washout angle is 0°, the lift-to-drag ratio reaches its highest point at a 3° angle of attack which is 16.65, and then the ratio starts to drop to 3.42 at an 18° angle of attack. When the washout angle is 5°, the lift-to-drag-ratio reaches its highest point at a 3° angle of attack which is 16.65, and then starts to decline to 3.47 at an 18° angle of attack. As the angle of attack increases, the airflow on the upper surface of the wing gradually moves to the front end. At an angle of attack of 9°, a small amount of airflow separation first appears in the middle and rear of the upper surface of the wing. As the angle of attack continues to increase, the airflow separation then spreads to both sides, and at 15°, the upper surface of the wing root is almost completely separated; as the angle of attack increases, the lift and drag of the wing increase, and the lift-to-drag ratio increases first and then decreases. When the angle of attack increases to a certain value, the lift reaches the maximum value and no longer increases. When the wing root twist angle is changed, the wing tip airflow separation gradually disappears as the angle of attack increases. When the twist angle is 5°, no airflow separation occurs on the wing tip plane, which effectively improves the wing stall effect; compared to no washout angle, at the same angle of attack, both lift and drag decrease, but the lift-to-drag ratio increases slightly. The wing has the largest lift-to-drag ratio at an angle of attack of 3-6 degrees.

The most important inspection is on the trend of the relationship between boundary layer separation (stall) and the washout angle. For an airfoil with a low washout angle (0°), the airfoil, especially at the wing tip, is prone to stalling at a relatively lower angle of attack. For instance, the airfoil with 0° washout and 7m from the symmetric plane is already stalling at an angle of attack of 15°, which compares to the airfoil at the wing root, which hasn’t stalled at an even higher angle of attack of 18°. Next, the critical angle appears to be increasing as the washout angle increases. For instance, the airfoil with 0° washout at 10m from the symmetrical plane is completely stalled at an angle of attack of 18°, which is significantly different from the result of the airfoil with 5° washout at 10m from the symmetrical plane, in which the boundary layer isn’t separated from the airfoil and isn’t stalled. This could be explained by the angle of incidence (AOI) of the airfoil. With the washout angle, the angle of incidence of the wing tip is lower than the wing root. So, when the wing roots are at the same angle of attack, the wing tips of wing foils with a higher washout angle will have a lower angle of attack, which delays the stall,

because a stall happens when the angle of attack is too high. Delaying the stall at the wing tip is extremely crucial because the aileron is located at the wing tip. This means that since the wing tip stalls later, the aileron is still able to maintain its function while the wing root has stalled, thus allowing the plane to be still controllable.

The empirical data in the data table showed relationships between the coefficient of lift and the coefficient of drag with the angle of attack and the washout angle. When the washout angles are the same, the coefficient of lift increases at a decreasing rate while the coefficient of drag increases at an increasing rate as the angle of attack increases. This is shown by the chart made in the previous section, which shows the curve of the coefficient of lift as concave down, while the curve for the coefficient of drag is concave up.

By comparing the data across different washout angles, it appears that a higher washout angle will lead to a lower lift coefficient and a lower drag coefficient at the same angle of attack. This is shown by the chart in the previous section. Additionally, the lift-to-drag ratio of airfoils with lower washout angle is higher than that of airfoils with higher washout angle at an angle of attack from 0°-3°. When the angle of attack exceeds 3°, the lift-to-drag ratio of airfoils with a higher washout angle is higher.

Conclusion:

To explore the effect of wing washout on the aerodynamic characteristics of the wing, a Boeing 737 wing model was used in this experiment. By changing the washout angle, models with washout angles of 0°, 3°, and 5° were established. The three-dimensional flow field of the wing was numerically simulated based on the SST k-omega turbulence model. The wing modeling and meshing were completed using ANSYS Fluent Student software, and the aerodynamic characteristics of the wing at different angles of attack under different washout angles were compared. The results show that appropriate wing washout can reduce the boundary layer separation at the wing tip at a high angle of attack and reduce the stall at the wing tip; as the washout angle increases, both the lift and drag coefficient of the wing decrease, but the lift-to-drag ratio rises slightly, and the lift-to-drag ratio is the largest at an angle of attack of 3°-5°. The result matches our hypothesis.

Aircraft engineers could utilize our results to pick the best washout angle that fits their needs on the performance of the airplane. This will increase the efficiency and safety of commercial aircraft.

We weren’t able to find the exact range of washout angles where the benefits of it are greater than the drawbacks of it. So, our experiment could have tested more washout angles and a wider range of angles of attack, which could increase the accuracy of our result.

Bibliography:

[1] **Andersen, D. (n.d.). *Understanding Washout* | *Model Aviation*. | Model Aviation. Retrieved November 29, 2024, from https://www.modelaviation.com/washout**

Washout is when the angle of a wing changes from the root (near the plane's body) to the tip, with the tip being slightly lower. This helps keep the plane more stable by making sure the root stalls (loses lift) first, so the tips keep working and the plane stays easier to control. It's used in full-size airplanes and model planes to prevent crashes, especially during slow flying or sharp turns. When building a model plane, you need to carefully twist the wing to get the right amount of washout.

[2] **Andersen, D. (n.d.). *Washout Why? How much and where*. Anderson Designs. Retrieved November 29, 2024, from https://www.rcscalebuilder.com/tutorials/washout/washout.pdf**

Washout is a twist in airplane wings that makes the tips hit the airflow at a lower angle than the root. It helps prevent stalls, keeps the plane stable, and improves control, especially at low speeds. The amount of washout depends on the wing shape and type of plane. It's important for some planes but not needed for aerobatic ones or planes flying upside down often. Builders can add washout during or after construction to improve flight performance.

[3] **C., W. (2024, July). *Aerodynamic characteristics of a pitching airfoil with leading-edge morphing*. Chinese Journal of Aeronautics. Retrieved November, 2024, from https://doi.org/10.1016/j.cja.2024.03.036**

This study looks at how changing the shape of the front edge (leading edge) of an airfoil can stop it from shaking too much, which happens when air flows break apart at high angles. Using computer simulations, the researchers found that if the front edge moves with a phase offset of 315°, it helps keep the airfoil stable by creating air vortices earlier

and pushing it down at the right time. This method could make airplanes and turbines more stable and efficient.

[4] **Cutler, C. (2024, September 14). *How Wing Washout Makes Your Airplane More Stable*. Boldmethod. Retrieved November 29, 2024, from https://www.boldmethod.com/learn-to-fly/aircraft-systems/wing-washout-makes-your-airplane-and-wings-more-stable-when-flying/**

Wing washout is when the angle of attack on an airplane's wing decreases from the root to the tip. This helps improve stability by preventing the entire wing from stalling at once. It makes the plane more stable during turns, especially at low speeds. While washout is great for general stability, too much can reduce performance in aerobatic maneuvers. It's used based on the aircraft type, with careful adjustments to achieve the best balance.

[5] **Whomsley, J. (2014, March 11). *Flying Wing Set-Up - Wash-Out and Reflex.* FLITE TEST. Retrieved November 29, 2024, from https://www.flitetest.com/articles/flying-wing-set-up-wash-out-and-reflex**

The article discusses the importance of wash-out for flying wings to ensure stability, especially during landings. Wash-out is a twist in the wing that causes the wingtips to have a lower angle of attack than the root, which helps prevent tip stalls. The author suggests that around 3-4 degrees of wash-out is typically enough and shares their experience of modifying their wings by twisting them to add wash-out for better control and stability.

[6] **Yılmaz, M., Koten, H., Çetinkaya, E., & Coşar, Z. (2018, December 31). *Journal of Energy Systems A comparative CFD analysis of NACA0012 and NACA4412 airfoils*. DergiPark. Retrieved November 29, 2024, from https://dergipark.org.tr/en/download/article-file/540472**

This article focuses on how the shape of airfoils (wing designs) impacts the performance of wind turbines. It discusses the use of Computational Fluid Dynamics (CFD) to analyze

various airfoil profiles like NACA 4420, NACA 0012, and others, studying how changes in angles of attack and Reynolds numbers affect lift, drag, and airflow. Researchers use CFD to simulate real-world conditions, saving time and costs compared to physical testing. Different turbulence models are applied in these studies to optimize wind turbine efficiency.